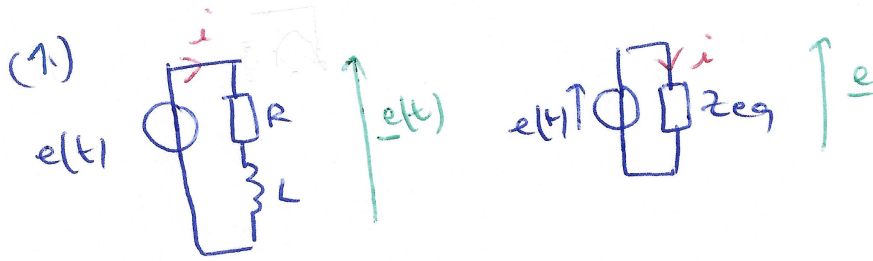


Régime variable

A. $e(t) = E \cos(\omega t - \frac{\pi}{3})$

$$\underline{e} = E e^{j\omega t} e^{-j\frac{\pi}{3}}$$



$$\underline{e} = \underline{z}_{eq} \underline{i}$$

$$\underline{i} = \frac{\underline{e}}{\underline{z}_{eq}}$$

$$\underline{z}_{eq} = \underline{z}_R + \underline{z}_L = R + j\omega L$$

$$\underline{i} = \frac{\underline{e}}{R + j\omega L}$$

(2) $i(t) = \text{Re}(\underline{i}) = I_0 \cos(\omega t + \varphi)$

$$\underline{i} = I e^{j\omega t} e^{j\varphi}$$

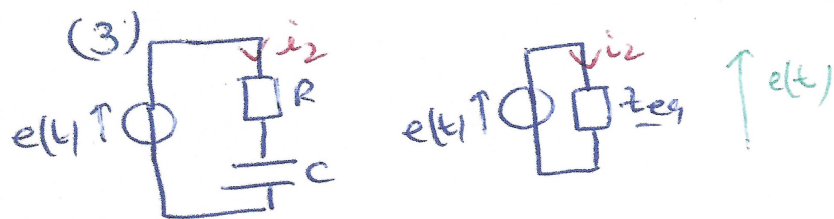
$$I = |\underline{i}| = \frac{|\underline{e}|}{|R + j\omega L|} = \frac{E}{\sqrt{R^2 + \omega^2 L^2}}$$

$$\begin{aligned} \arg\left(\frac{\underline{i}}{\underline{e}}\right) &= \varphi + \frac{\pi}{3} = \arg\left(\frac{1}{R + j\omega L}\right) \\ &= -\arg(R + j\omega L) \\ &= -\text{Arctan} \frac{\omega L}{R} \end{aligned}$$

$$\begin{aligned} \frac{\underline{i}}{\underline{e}} &= \frac{I e^{j\omega t} e^{j\varphi}}{E e^{j\omega t} e^{-j\pi/3}} \\ &= \frac{I}{E} e^{j(\varphi + \pi/3)} \end{aligned}$$

3 $\Rightarrow \varphi = -\frac{\pi}{3} - \text{Arctan} \frac{\omega L}{R}$

Final
$$i(t) = \frac{E}{\sqrt{R^2 + \omega^2 L^2}} \cos\left(\omega t - \frac{\pi}{3} - \text{Arctan} \frac{\omega L}{R}\right)$$



$$e(t) = z_{eq} \underline{i}_2$$

$$\underline{i}_2(t) = \frac{e(t)}{z_{eq}}$$

$$z_{eq} = R + \frac{1}{j\omega C}$$

$$\underline{i}_2(t) = \frac{e}{R + \frac{1}{j\omega C}}$$

$$\underline{i}_2 = I_2 e^{j\omega t} e^{j\varphi}$$

$$= \frac{j\omega C e}{1 + jRC\omega}$$

$$(4) |\underline{i}_2| = \frac{|e|}{|R + \frac{1}{j\omega C}|} = \frac{E}{\sqrt{R^2 + \frac{1}{C^2\omega^2}}}$$

$$\arg\left(\frac{\underline{i}_2}{e}\right) = \varphi + \pi/3$$

$$= \arg\left(\frac{1}{R + \frac{1}{j\omega C}}\right)$$

$$= -\arg\left(R + \frac{1}{j\omega C}\right) = -\arg\left(R - \frac{1}{\omega C}j\right)$$

$$= -\text{Arctan}\left(\frac{-1/\omega C}{R}\right)$$

$$= +\text{Arctan}\frac{1}{RC\omega}$$

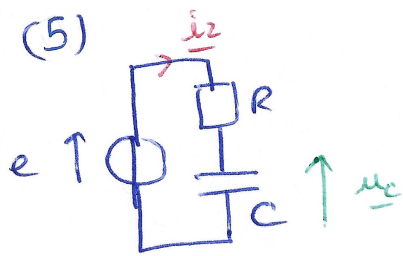
$$= +\left(\frac{\pi}{2} - \text{Arctan} RC\omega\right) = \frac{\pi}{2} - \text{Arctan} RC\omega$$

$$\begin{aligned} \frac{\underline{i}_2}{e} &= \frac{I_2 e^{j\omega t} e^{j\varphi}}{E e^{j\omega t} e^{-j\pi/3}} \\ &= \frac{I_2}{E} e^{j(\varphi + \pi/3)} \end{aligned}$$

$$\Rightarrow \varphi = -\pi/3 + \pi/2 - \text{Arctan} RC\omega = \pi/6 - \text{Arctan} RC\omega$$

$$\underline{i}_2(t) = \frac{E}{\sqrt{R^2 + \frac{1}{C^2\omega^2}}} \cos\left(\omega t - \frac{\pi}{3} + \text{Arctan} \frac{1}{RC\omega}\right)$$

$$\text{ou } i_2(t) = \frac{E}{\sqrt{R^2 + \frac{1}{C^2 \omega^2}}} \cos \left(\omega t + \frac{\pi}{6} - \text{Arctan}(RC\omega) \right).$$



1^{ère} méthode $\underline{u}_c = \underline{z}_c \underline{i}_2$

$$= \frac{1}{jC\omega} \underline{i}_2$$

$$= \frac{1}{jC\omega} \frac{\underline{e}}{R + \frac{1}{jC\omega}}$$

$$\underline{u}_c = U e^{j\omega t + j\varphi'}$$

$$\underline{e} = E e^{j\omega t - j\pi/3}$$

$$\underline{u}_c = \frac{\underline{e}}{1 + jRC\omega}$$

(6) $\underline{u}_c(t)$.

$$|\underline{u}_c| = U = \frac{|\underline{e}|}{|1 + jRC\omega|} = \frac{E}{\sqrt{1 + (RC\omega)^2}}$$

$$\arg \left(\frac{\underline{u}_c}{\underline{e}} \right) = \varphi' + \pi/3 = \arg \frac{1}{1 + jRC\omega} = -\arg(1 + jRC\omega) = -\text{Arctan}(RC\omega).$$

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$$\underline{u}_c(t) = \frac{E}{\sqrt{1 + (RC\omega)^2}} \cos \left(\omega t - \pi/3 - \text{Arctan}(RC\omega) \right).$$

(2^{ème} méthode: pont diviseur de tension.)